black holes always have apparent horizons), *every black hole must have a singularity inside itself.*

The most amazing thing about this *singularity theorem* was its sweeping power. It dealt not solely with idealized imploding stars that have special, idealized properties (such as being precisely spherical or having no pressure); and it dealt not solely with stars whose initial random deformations are tiny. Instead, it dealt with every imploding star imaginable, and thus, undoubtedly, with the real imploding stars that inhabit our real Universe.

Penrose's singularity theorem acquired its amazing power from a new mathematical tool that he used in its proof, a tool that no physicist had ever before used in calculations about curved spacetime, that is, in general relativistic calculations: *topology.*

Topology is a branch of mathematics that deals with the qualitative ways in which things are connected to each other or to themselves. For example, a coffee cup and a doughnut "have the same topology" because (if they are both made from putty) we can smoothly and continuously deform one into the other without tearing it, that is, without changing any connections (Figure 13.5a). By contrast, a sphere has a different topology from a doughnut; to deform a sphere into a doughnut, we must tear a hole in it, thereby changing how it is connected to itself (Figure 13.5b).

Topology cares *only* about connections, and *not* about shapes or sizes or curvatures. For example, the doughnut and the coffee cup have very different shapes and curvatures, but they have the same topology.

We physicists, before Penrose's singularity theorem, ignored topology because we were fixated on the fact that spacetime *curvature* is the central concept of general relativity, and topology cannot tell us anything about curvature. (Indeed, because Penrose's theorem was based so strongly on topology, it told us nothing about the singularity's curvature, that is, nothing about the details of its tidal gravity. The theorem simply told us that somewhere inside the black hole, spacetime comes to an end, and anything that reaches that end gets destroyed. *How* it gets destroyed was the province of curvature; *that* it gets destroyed-that there is an end to spacetime—was the province of topology.)

If we physicists, before Penrose, had only looked beyond the issue of curvature, we would have realized that relativity *does* deal with questions of topology, questions such as "Does spacetime come to an end (does it have